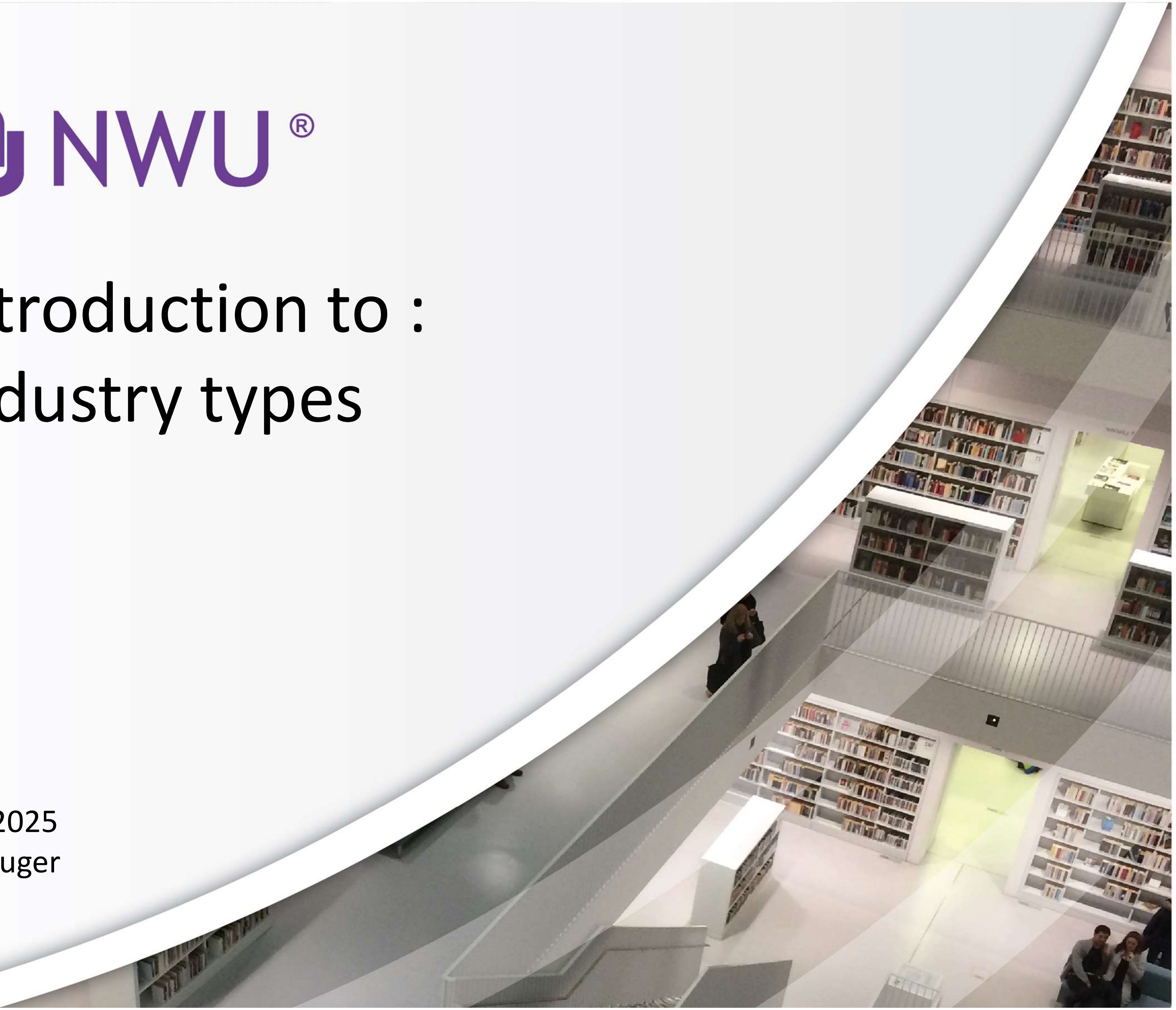
Introduction to : Industry types



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W Kruger

Different
Industries

Different
Purposes

Different
Needs

Different
Designs

Products – transport (airplanes, trains, boats, cars, bikes, bicycles) clothes, food(fresh, cooked, tinned), Equipment (kitchen eg. kettle, oven, washing machine, office eg. computers, printers, stationary) work eg artisan tools, sport eg rackets, balls, hockey sticks), packaging, books, toys etc.

Services – restaurants, cafés, shopping malls, stadiums, events, airports, taxi ranks, administration offices, packaging, cooling, storing facilities, parking lots, schools, universities, homes, flats, sports etc.

Let's have a look at some... *Notice*

- Differences in flows (materials, people and info)
- Interactions of machines vs people
- Acceptance criteria (what is the critical factors for THAT facility)
- How this affects Facility Design

Product manufacturing - Cars

Product manufacturing - Cellphones

Product manufacturing - Food

Services - Hospital

Services - Airport

Services Theme park

Building vs Rebuilding a facility

From scratch = *Green field*

eg. a new home, a new store, a new plant, a new site

Upgrading or redefining a facility = *Brown field*

eg. When a store is sold to another company, when a place need to be revamped or repurposed, improved

FACIILITIES DESIGN is always the big money – that is why the DESIGN is so critical

A Cost example

- Let’s say average construction cost = R10 000/sqm
- Determine the cost of this lecture hall
- $R10000 \times x = R$__________

A facility affects:

- Operating costs
- Operational effectiveness
- Handling and maintenance costs
- Productivity
- Employee morale
- Flexibility to adapt to customer requirements
- Sustainability over the longterm

So some BIG implications!

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Some characteristics of designs

- Flexible (if possible)
- Modular
- Upgradable
- Adaptable
- Scalable
- Repeatable
- Environment and energy friendly

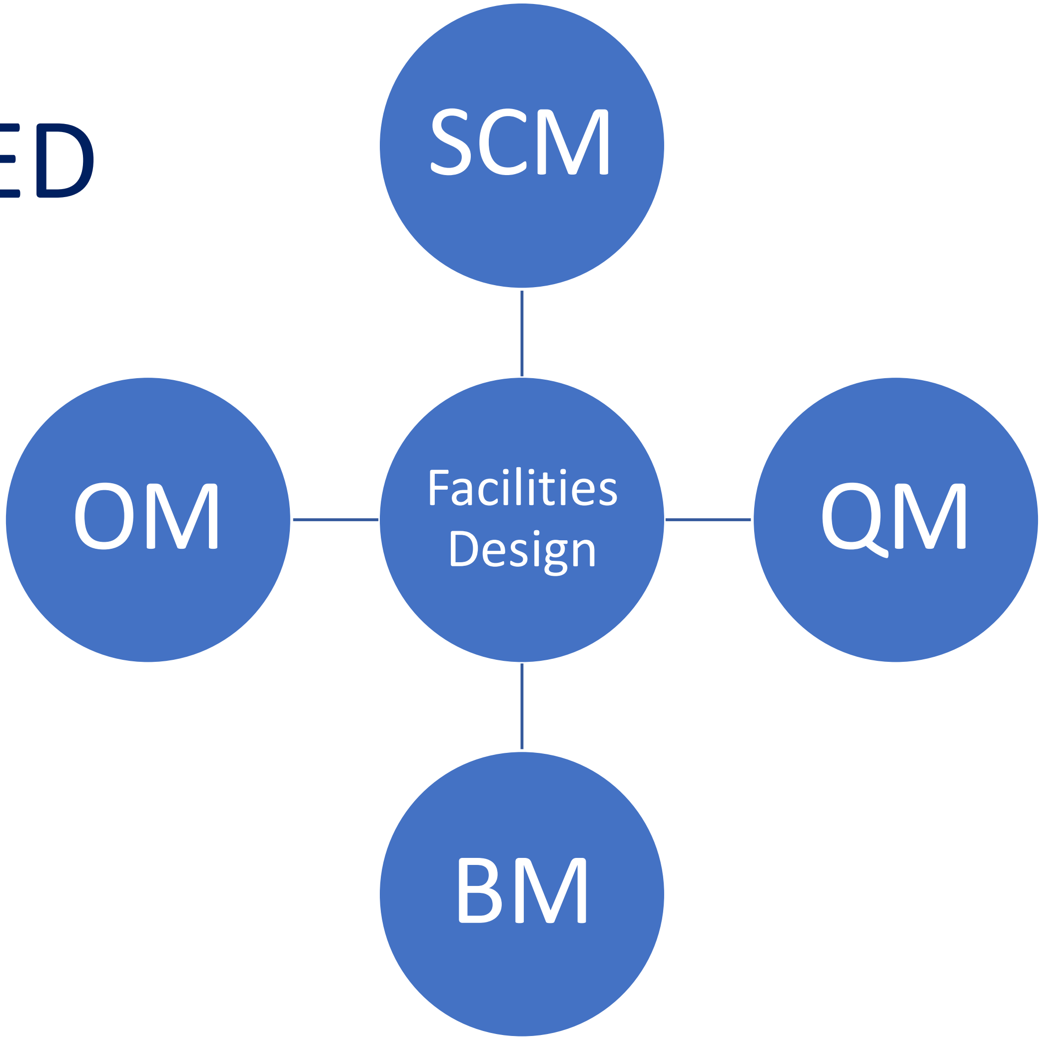
6 key questions to answer before Starting a design…

1. WHAT is produced / or service to be given?
2. HOW …
3. HOW MUCH… (capacity planning)
4. WHERE ….
5. WHEN…
6. For HOW LONG…

Refer Ch2 textbook

Facilities design is an

INTEGRATED course



In manufacturing the specifics include:

Product specs

Process specs

Production Schedule

A	B	C
x		x
	x	
		x

Scheduling specs

Product design tools

- Quality Function Deployment (QFD)
- Benchmarking
- Prototyping
- Assembly drawings
- Photos

Process design tools

- Bill of Materials (BOM)
- Route sheets
- Assembly chart
- Operation process flow charts
- Precedence diagrams
- Material Requirements Planning (MRP)

Management and Planning tools

1. Affinity diagram
2. Interrelationship graphs
3. Tree diagrams
4. Prioritization matrices
5. Matrix diagrams
6. Process decision program chart
7. Activity network diagram

Refer textbook
Fig 2.26 p73

also Process Flow charts or
Value Stream Maps (VSM)

Bottom line of all tools….

To help you have
ENOUGH
ACCURATE
DATA
to do calculations and work with